

Test Results — AuthShield 360 v2

Every number in this report is **measured** from a real run of the four suites on a pristine database, and the raw log of each suite is kept in `docs/technical/Test_Run_Logs/`.

1) Verdict

ALL GREEN — 140 checks · 0 failures

- Run window: `2026-09-28T18:41:47.367Z` → `2026-09-28T18:42:55.293Z`
- Node version: `v24.19.0` · server: `http://localhost:4000`
- Database rebuilt with `node scripts/reset-db.js` before the run (reproducibility requirement).
- Reproduction: `node scripts/run-full-test-suite.js`

2) Suite details

#	Suite	Checks	Passed	Failed	Time	Exit code	Raw log
1	Risk engine + core auth (no server)	17	17	0	6s	0	01-smoke.log
2	Full user journey on a live server	77	77	0	33.7s	0	02-e2e.log
3	UI in a real browser	28	28	0	14.9s	0	03-ui.log
4	Data resilience after process restart (NFR)	18	18	0	11.1s	0	04-restart.log
	Total	140	140	0			

3) Requirements coverage

Requirement	How covered	Evidence
Adaptive risk scoring	e2e: low/medium/high-risk logins	02-e2e.log
Multi-factor authentication	e2e: STEP_UP then valid/invalid OTP	02-e2e.log
RBAC for student/teacher/admin	e2e: school portal + admin-route denial	02-e2e.log
Brute-force (failed-login) protection	e2e: failure streak + lockout	02-e2e.log
Account recovery / reset	e2e: RECOVERY_REQUEST → RESET	02-e2e.log
Digital forensics	e2e: open case, capture evidence, export	02-e2e.log
Attack simulation A/B/C	e2e: SME + SOC	02-e2e.log

Requirement	How covered	Evidence
Scenario benchmarking	e2e: 3 scenarios from real telemetry	<code>02-e2e.log</code>
SSE live streaming	e2e: audit/alert events	<code>02-e2e.log</code>
UI & themes	ui: 28 checks in a real browser	<code>03-ui.log</code>
NFR: data survives restart	restart: 18 checks on :4001	<code>04-restart.log</code>

4) RBAC evidence from the database

Metric	Value
HIGH_RISK_BLOCK	9
FAILED_PASSWORD	8
STEP_UP	6
INVALID_OTP	2
Admin-route breach attempts (UNAUTHORIZED_ROUTE)	1
Logins from a blacklisted IP	0
IMPOSSIBLE_TRAVEL signals	0

5) Brute-force protection evidence

Mechanism	Implementation	Evidence
Consecutive-failure counter	<code>resetStreak()</code> + CONSECUTIVE_FAILURE factor (weight 20)	8 failed attempts logged
Lockout window	<code>threat_factors</code> with short TTL (5 min by default)	<code>threat_factors</code> table
OTP rejection	<code>verifyTotp()</code> before issuing a session	2 rejections
High-risk block	<code>HIGH_RISK_BLOCK</code> at score ≥ 70	9 blocks
SOC alert	<code>flagAlert()</code> on every block / step-up	14 alerts

6) Real risk factors in the log

Risk factors	Frequency
BASELINE	12
ACCOUNT_LOCKED	6
NEW_IP + UNRECOGNIZED_DEVICE	3
CROSS_DEVICE_PENDING + NEW_IP + UNRECOGNIZED_DEVICE	3
UNRECOGNIZED_DEVICE	2
TRUSTED_DEVICE_APPROVAL	2
NEW_IP	2
NEW_IP + UNRECOGNIZED_DEVICE + CONSECUTIVE_FAILURES	2

7) Forensics & simulation evidence

Table	Rows
cases	1
case_items (linked evidence)	45
attack_runs (attack simulations)	3
alerts (SOC alerts)	14
audit_trail (all attempts)	41

8) Authentication scenarios as exercised by the tests

Scenario	Attempts	Succeeded	Rejected / suspended
1 — password only	18	4	14
2 — + Mobile OTP	13	4	9
3 — + email and RBAC	5	3	2

Security / usability comparison details: [Scenario_Comparison_EN.md](#).

Full authentication logs: [Authentication_Logs_EN.md](#).